

Table 2: Adaptation Conditions

Condition	n_{a_E}	n_{a_I}	n_{b_E}	n_{b_I}	Description
No Adaptation	0	0	0	0	Baseline
SFA Only	3	0	0	0	Spike-frequency adaptation enabled
STD Only	0	0	1	0	Short-term depression enabled
SFA + STD	3	0	1	0	Both mechanisms enabled

Effect on parameters:

- When $n_{a_E} = 0$: No SFA variables a_{ik} ; the $c_E \sum_k a_{ik}$ term is zero.
- When $n_{a_E} = 3$: Three SFA timescales with $\tau_{a_k} \in \{0.25, 1.58, 10\}$ s (logspaced) and coupling $c_E = \frac{0.5}{3}$.
- When $n_{b_E} = 0$: No STD variable b_i ; synaptic output equals r_i (equivalent to $b_i = 1$).
- When $n_{b_E} = 1$: STD enabled with a single timescale, recovery constant $\tau_{rec} = 1$ s and release constant $\tau_{rel} = \frac{1}{4}$ s; the resource enters the recurrent term as the product $b_i r_i$.

Inhibitory neurons have no adaptation mechanisms ($n_{a_I} = n_{b_I} = 0$).

Implementation note: When any of n_{a_E} , n_{a_I} , n_{b_E} , or n_{b_I} is set to zero, the corresponding state variables (a_{ik} or b_i) are excluded from the system state vector and the Jacobian matrix. This prevents spurious zero eigenvalues that would otherwise arise from including disabled adaptation dynamics.